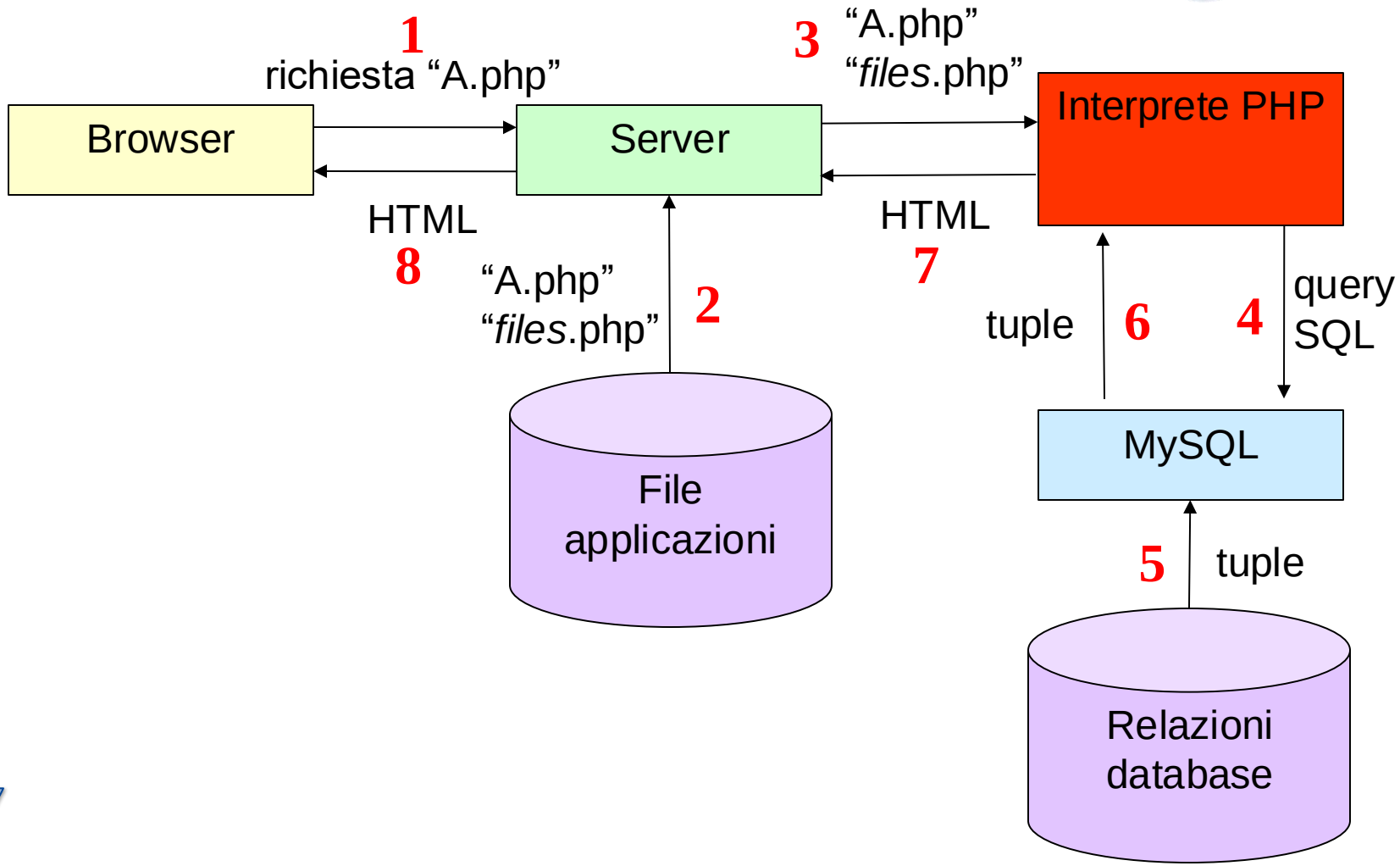
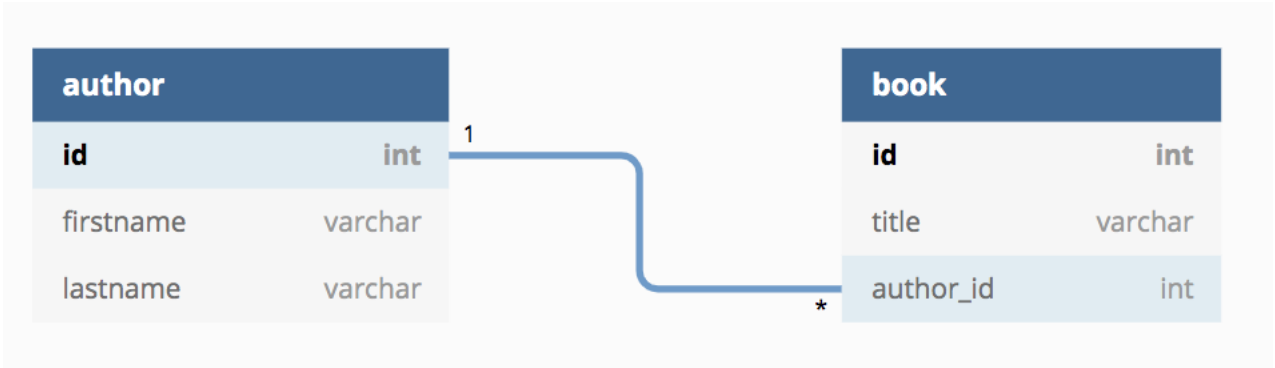
PROGRAMMAZIONE WEB E SERVIZI DIGITALI

PHP E STRUMENTI DI PERSISTENZA (MYSQL)

Interazione tra PHP e MySQL



Il database


author
id int
firstname varchar
lastname varchar
1
*
book
id int
title varchar
author_id int

Connessione a MySQL


```php
new PDO("mysql:host={$HOST};dbname={$DB_NAME}", $USERNAME, $PASSWORD);
```

PHP Data Object - apre una connessione al server ***$HOST***, database ***$DB_NAME***, con utente ***$USERNAME*** e password ***$PASSWORD***

Esempio (I)


```php
<?php
    $USERNAME = "devis";
    $PASSWORD = "bianchini";
    $HOST = "localhost";
    $DB_NAME = "MyLibraryDB";
```

utils/config.php

Esempio (II)


```php
include_once('../utils/config.php');
include_once('Author.php');
include_once('Book.php');

2 references | 0 implementations
class DataLayer
{
    4 references
    private $pdo;

    /**
     * Constructor of the DataLayer class
     */
    2 references | 0 overrides
    public function __construct()
    {
        global $HOST, $USERNAME, $PASSWORD, $DB_NAME;

        try {
            $this->pdo = new PDO("mysql:host={$HOST};dbname={$DB_NAME}", $USERNAME, $PASSWORD);
            // Set the PDO error mode to exception
            $this->pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
        } catch(PDOException $e){
            die("ERROR: Could not connect. " . $e->getMessage());
        }
    }
```

Esecuzione di query SQL

`(new PDO)->query($sql)`:
invia una query SQL al database a cui si è attualmente connessi

a)

```php
$sql = "SELECT * FROM book ORDER BY title";
try {
    $statement = $this->pdo->query($sql);
```

b)

```php
$sql = "SELECT * FROM author where id = :id";
try {
    $statement = $this->pdo->prepare($sql);
    $statement->execute(array(':id' => $id));
```

c)

```php
$sql = "SELECT * FROM author where id = :id";
try {
    $statement = $this->pdo->prepare($sql);
    $statement->bindParam(':id',$id);
    $statement->execute();
```

Elaborazione del risultato (I)

`$queryStatement->fetch(PDO::FETCH_ASSOC)`:
estrae un record dal risultato di una query (ogni volta che viene invocata) sotto forma di array associativo (i nomi delle colonne del record diventano i nomi degli elementi dell' array)

```php
public function findAuthorById($id)
{
    $sql = "SELECT * FROM author where id = :id";
    try {
        $statement = $this->pdo->prepare($sql);
        $statement->execute(array(':id' => $id));

        $author = $statement->fetch(PDO::FETCH_ASSOC);

        // Check if author found
        if ($author) {
            return new Author($author['id'], $author['firstname'], $author['lastname']);
        } else {
            return null; // Author not found
        }
    } catch (PDOException $e) {
        die("ERROR: Could not execute query. " . $e->getMessage());
    }
}
```

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{
    $sql = "SELECT * FROM author where id = :id";
    try {
        $statement = $this->pdo->prepare($sql);
        $statement->execute(array(':id' => $id));

        $author = $statement->fetch(PDO::FETCH_ASSOC);

        // Check if author found
        if ($author) {
            return new Author($author['id'], $author['firstname'], $author['lastname']);
        } else {
            return null; // Author not found
        }
    } catch (PDOException $e) {
        die("ERROR: Could not execute query. " . $e->getMessage());
    }
}
```

$autho

id	2
firstname	Alessandro
lastname	Manzoni

Elaborazione del risultato (II)

`$queryStatement->fetchAll(PDO::FETCH_ASSOC)`: simile al precedente, ma in questo caso recupera tutti i record nel risultato di una query, mettendoli in un array di array associativi

```php
public function listBooks()
{
    $sql = "SELECT * FROM book ORDER BY title";
    try {
        $statement = $this->pdo->query($sql);

        $books = $statement->fetchAll(PDO::FETCH_ASSOC);
        $booksList = array();
        foreach ($books as $book) {
            $author = $this->findAuthorById($book['author_id']);
            $booksList[] = new Book($book['id'],$book['title'],$author->getFirstName()." ".$author->getLastName(),$book['author_id']);
        }
        return $booksList;
    } catch (PDOException $e) {
        die("ERROR: Could not execute query. " . $e->getMessage());
    }
}
```

Modifica del database

`$queryStatement->rowCount()`:
restituisce il numero di tuple coinvolte nella query appena eseguita

```php
public function findBooksByAuthorID($author_id)
{
    $sql = "SELECT * FROM book where author_id = :author_id";
    try {
        $statement = $this->pdo->prepare($sql);
        $statement->bindParam(':author_id',$author_id);
        $statement->execute();

        // Get the number of affected rows
        $affectedRows = $statement->rowCount();

        if($affectedRows != 0)
        {
            return true;
        } else {
            return false;
        }
    } catch (PDOException $e) {
        die("ERROR: Could not execute query. " . $e->getMessage());
    }
}
```

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